SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Nighil Adhithyar *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	1	1
Application & Analysis	20%	1	1	2
Explanation & Clarity	15%	0	1	0

Total Marks 21 **/30**

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

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2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

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Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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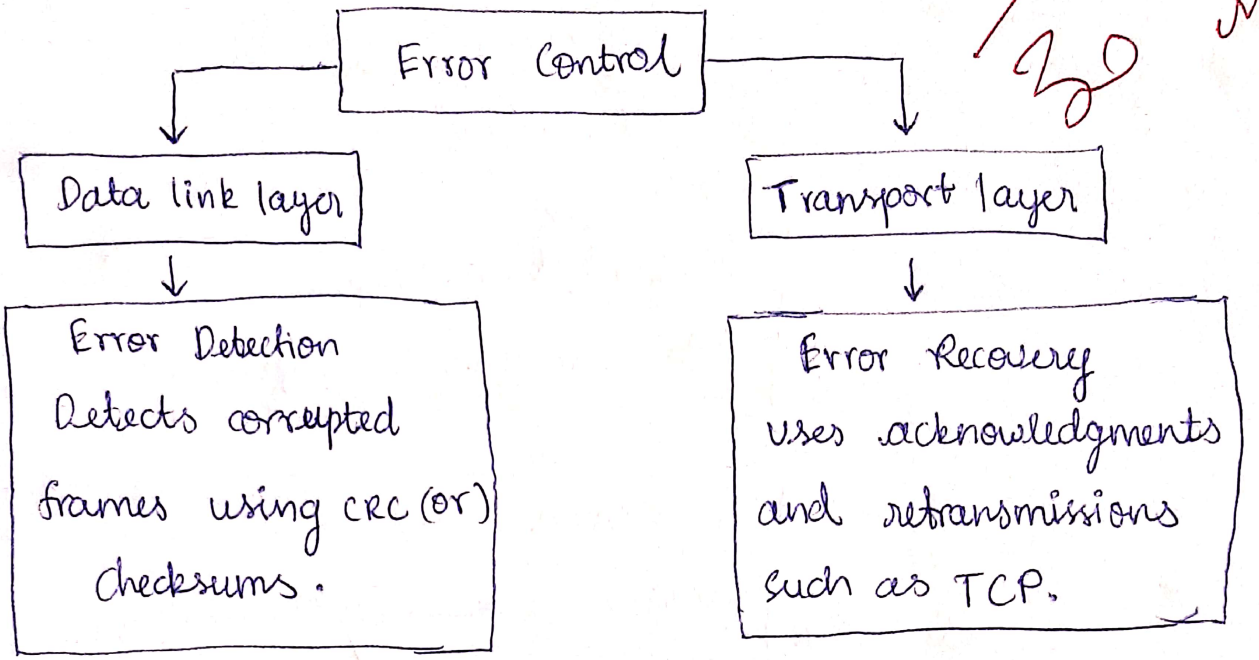
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CSA 0710

Computer networks : Concept Map

Nigil Adhithya R
(191126026T) 26

4) Error control in data transmission



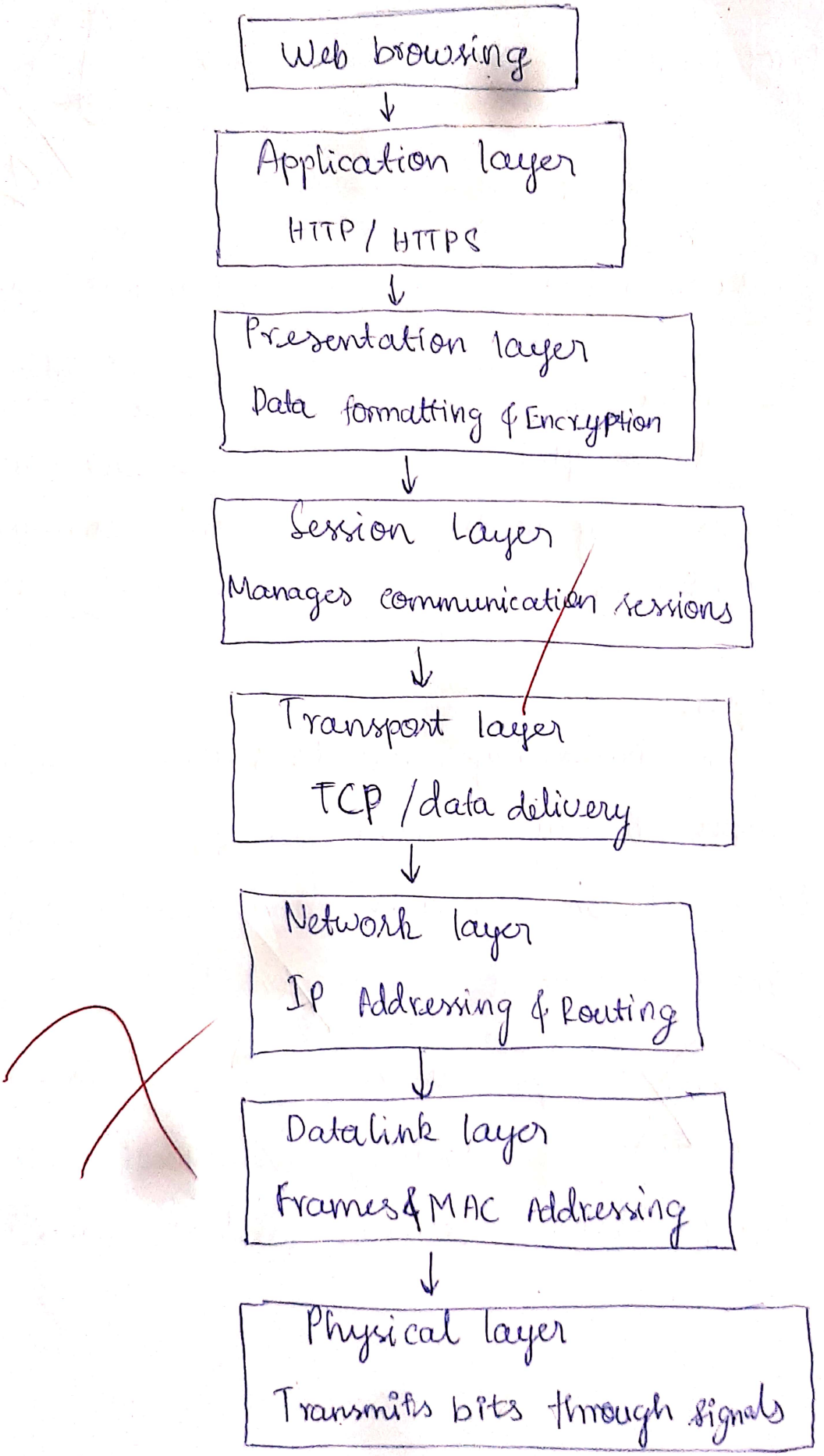
Analysis:

The datalink layer detects Errors in frames transmitted between directly connected devices. If a frame is corrupted, Error detection mechanisms can identify the problem, & some link protocol can request transmission.

The transport layer, Particularly TCP, provides end-to-end reliability by detecting missing or corrupted data, using acknowledgements and retransmitting lost signals

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5) concept map for web browsing using OSI Layers.



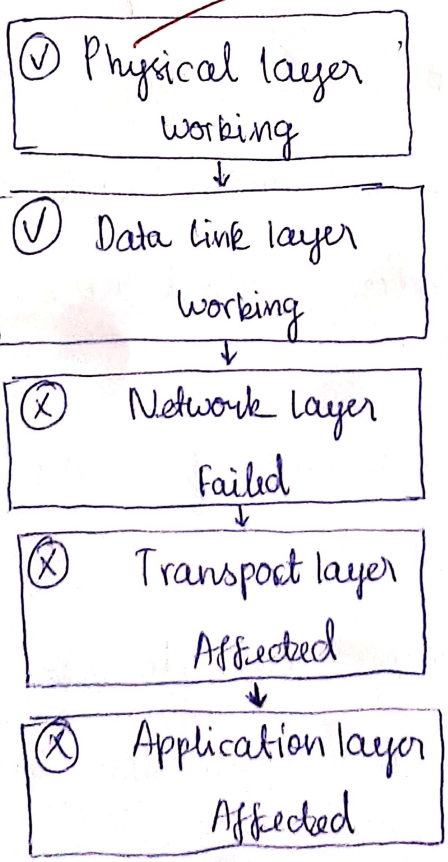
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Analysis:

Failure in any one layer can interrupt communication. For example, A DNS or app problem may prevent a website from loading, a routing failure may prevent packets from reaching the server if a physical connection failure may disconnect the device. Therefore every layer should be functioning properly for successful web browsing.

6) Troubleshooting Using the OSI concept Map.

Concept Map:



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Analysis :

The physical & Data Link layer are working correctly, Indicating that the device has a physical connection and local link communication communication is functioning. However the Network layer has failed, Possibly due to an incorrect IP Address, a routing problem or a gateway configuration issue.

Because packets can't reach their destination, the transport layer cannot establish or maintain end-to-end communication, and applications such as web browsers may fail to connect.

How it helps in troubleshooting :-

- Identify the first failed layer - the Network layer
- Check IP Address, subnet mask, default gateway & Routing.
- Test the connectivity using commands such as Ping & tracert.
- After fixing the network problem, test the transport and application layers again.

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